SECTION A [25 *marks*]

This section consists of **3** questions. Answer **all** questions.

1 Evaluate the following limits, if they exist

(a) $\lim\limits_{x \to -\infty} \sqrt{\dfrac{3x+1}{6x-3}}$ [4 *marks*]

(b) $\lim\limits_{x \to 0} \dfrac{x-2x^2}{1-\sqrt{x+1}}$ [5 *marks*]

2 Find $\dfrac{dy}{dx}$ of the following

(a) $y = \dfrac{3e^{2x}}{x^2-1}$ [5 *marks*]

(b) $y = \sqrt{x^3-5x} + \ln\left(\dfrac{x+2}{2x-1}\right)$ [6 *marks*]

3 Given $f(x) = ax^3 + \dfrac{1}{2}x^2 - bx$ and its stationary points are $\left(-2, \dfrac{10}{3}\right)$ and $\left(1, -\dfrac{7}{6}\right)$.

Find the values of constant a and b. Hence, identify the extremum points.

[5 *marks*]

SECTION B [75 *marks*]

This section consists of **7** questions. Answer **all** questions.

1 Given complex number $i=\sqrt{-1}$, find the values of x and y if $\dfrac{1-i}{x-2yi}=\dfrac{1+i}{(2-i)^2}$.

Hence, express complex number $z=y+xi$ in polar form. [9 *marks*]

2 Solve

(a) $\sqrt{2\sqrt{x+1}}=\sqrt{3x-5}$ [3 *marks*]

(b) $11^{2x+1}\cdot 3^{x-2}=14^{x-1}$ [4 *marks*]

(c) $\dfrac{1}{|x-5|}\leq\dfrac{2}{|x|}$ [5 *marks*]

3 Given that $f(x)=\dfrac{ax+3}{x-b}$ and $g(x)=5-x$.

(a) Find the values of *a* and *b* if $f(2)=7$ and $f^{-1}(1)=-4$. [5 *marks*]

(b) Hence, determine the value of *c* such that $(g\circ f^{-1})(c)=\dfrac{1}{2}$. [3 *marks*]

4 The functions f and g are defined by $f(x)=1+2e^{x}$ and $g(x)=\ln\left(\dfrac{x-1}{2}\right)$.

(a) Show that $f(x)$ is one-to-one function. [2 *marks*]

(b) Find $(f\circ g)(x)$ and $(g\circ f)(x)$. Then, state the relation of f and g.

[4 *marks*]

(c) State the domain and range of f and g. [2 *marks*]

(d) On the same axes, sketch the graphs of f and its inverse. [3 *marks*]

5 (a) Given a piecewise function, $f(x)=\begin{cases} 2x+1 & , \quad x<-2 \\ 1-x^2 & , -2\le x<0 \\ 2e^{ax}+b & , \quad x\ge 0 \end{cases}$

where a and b are constant.

(i) Does $\lim\limits_{x\to -2} f(x)$ exist? Explain. [3 *marks*]

(ii) Find the values a and b if $f(x)$ is continuous at $x=0$ and $f'(1)=f'(0)$. [4 *marks*]

(b) A function $g(x)$ is defined by $g(x)=\dfrac{5x^2+8x+4}{x+1}$.

Compute its vertical and horizontal asymptotes. [4 *marks*]

6 (a) Let $x^2+y^2=2y$, find $\dfrac{dy}{dx}$. Show that $\dfrac{d^2y}{dx^2}=\dfrac{1}{(1-y)^3}$.

Then, get in term of y for $\left(\dfrac{2}{x}\right)\left(\dfrac{dy}{dx}\right)^3-x^2\left(\dfrac{d^2y}{dx^2}\right)$. [8 *marks*]

(b) Given that $x=e^t$ and $y=\dfrac{1}{1+e^t}$

(i) Show that $\dfrac{dy}{dx}<0$. [4 *marks*]

(ii) Find $\dfrac{d^2y}{dx^2}$ at $t=0$. [4 *marks*]

7 Salt is poured from a container at 10 $cm^3\ s^{-1}$ and it formed a conical pile whose height at any time is $\dfrac{1}{5}$ the radius of the base. At what rate is the height of the cone increasing when the height is 2 cm form the base of the cone ? [9 *marks*]

END OF QUESTION PAPER

Answers

Section A

1. (a) $\sqrt{\frac{1}{2}}$ (b) -2

2. (a) $\frac{dy}{dx}=\frac{6e^{2x}\left(x^2-x-1\right)}{\left(x^2-1\right)^2}$ (b) $\frac{dy}{dx}=\frac{3x^2-5}{2\sqrt{x^3-5x}}+\frac{1}{x+2}-\frac{2}{2x-1}$

3. $a=\frac{1}{3},\ b=2,\ \left(-2,\frac{10}{3}\right)$ maximum point, $\left(1,-\frac{7}{6}\right)$ minimum point

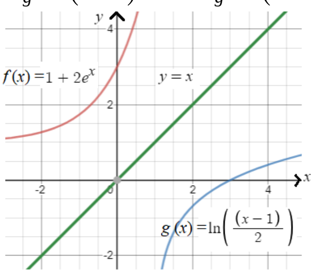
Section B

1. $x=-4,\ y=\frac{3}{2}, z=\frac{\sqrt{73}}{2}\left[\cos -1.21+i\sin -1.21\right]$

2. (a) $x=3$
   (b) $x=-0.872$
   (c) $\left\{x : x<0 \cup 0<x\le\frac{10}{3}\cup x\ge 10;\ x\ne 0,5\right\}$

3. (a) $a=2,\ b=1$
   (b) $c=\frac{24}{7}$

4. (a) Shown !
   (b) $(f\circ g)(x)=(g\circ f)(x)=x$, inverse of each others.
   (c) $D_f=(-\infty,\infty),\ R_f=(1,\infty)$
   $D_g=(1,\infty)\quad ,\ R_g=(-\infty,\infty)$
   (d)



5. (a) (i) $\lim_{x\to -2} f(x)$ exist, $\lim_{x\to -2} f(x)=-3$ (ii) $a=0,\ b=-1$
   (b) vertical asymptote: line $x=-1$
   horizontal asymptote: No

6. (a) $\frac{dy}{dx}=\frac{x}{1-y}$, Shown, $\frac{y(2-y)}{(1-y)^3}$
   (b) (i) Shown (ii) , $\frac{d^2y}{dx^2}=\frac{1}{4}$

7. $\frac{1}{10\pi}cms^{-1}$